Acoustic-Levitated Dynamic Voxel System for Volumetric Field Visualization

A Physical Application of Single Entity Field Interpretation (SEFI) for Microsoft Interactive AI Companions and Spatial Systems

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Abstract

This paper presents a dynamic volumetric rendering system that uses acoustic levitation to position and manipulate a single illuminated particle as a free-space voxel. By controlling the particle's trajectory through ultrasonic phased-array modulation, the system produces continuous three-dimensional curves and structures via persistence of vision. A geometric field model—specifically the Single Entity Field Interpretation (SEFI)—serves as the trajectory generator, mapping worldlines, curvature fields, and layered field constructs into real-time voxel motion. Illumination synchronized with particle position encodes additional geometric or physical attributes, enabling expressive volumetric representations within an open interaction volume.

The proposed architecture is low-cost, safe, and based entirely on established acoustic levitation techniques. It integrates a phased-array controller, a SEFI-driven coordinate engine, and a synchronized illumination subsystem to create a compact, research-grade instrument for field visualization and physical AI interaction. This approach offers a direct physical embodiment of geometric computation. The resulting platform supports scientific visualization, mixed-reality interaction, and physical AI companion interfaces, providing a practical foundation for future volumetric display research and Microsoft Copilot spatial interaction models.

1 Introduction

Volumetric displays aim to create perceivable three-dimensional structures in physical space, enabling users to observe geometry, motion, and field behavior without reliance on flat screens or head-mounted devices. Existing approaches are often constrained by limited viewing angles, high system complexity, environmental sensitivity, or significant operational cost. These limitations motivate the development of a simpler, safer, and more direct method for generating dynamic volumetric content.

Acoustic levitation provides a practical foundation for free-space voxel generation. Ultrasonic phased arrays can create stable pressure nodes capable of suspending small particles at precise coordinates within an open volume. By modulating the phase relationships across the array, the levitated particle can be repositioned rapidly and smoothly, enabling dynamic trajectories through three-dimensional space. When illuminated in sync with its motion, the particle acts as a controllable voxel whose path forms volumetric curves and structures through persistence of vision.

This paper introduces a volumetric rendering system that uses a single acoustically levitated particle as a dynamic voxel driven by a geometric field model. The Single Entity Field Interpretation (SEFI) provides the mathematical framework for generating worldlines, curvature fields, and layered field constructs. These geometric outputs are translated into real-time voxel trajectories, allowing the levitated particle to physically trace the structure of the underlying field model. Illumination synchronized with particle position encodes additional geometric or physical attributes, producing expressive volumetric representations within a compact and low-cost apparatus.

By combining established acoustic levitation techniques with a computational field model, this system offers a new class of volumetric visualization instrument: one that is inexpensive, safe, and capable of rendering dynamic geometric structures directly in free space for interactive AI platforms. The following sections describe the system architecture, levitation physics, rendering model, SEFI integration, control mechanisms, and feasibility considerations.

2 System Overview

The proposed volumetric rendering system uses acoustic levitation to position and manipulate a single illuminated particle as a dynamic voxel within an open three-dimensional interaction volume. The system is composed

of four coordinated subsystems: an ultrasonic phased array, a levitated particle, an illumination module, and a geometric field engine that generates real-time voxel trajectories. Together, these components form a compact, low-cost instrument capable of producing continuous volumetric curves and structures through persistence of vision.



Figure 1: Acoustic levitation hardware assembly featuring top and bottom ultrasonic transducer plates suspending a physical illuminated voxel particle in free space.

The ultrasonic phased array creates the acoustic field used for levitation and motion control. By adjusting the phase relationships across its transducers, the array forms stable pressure nodes at precise spatial coordinates. Dynamic modulation of these phase patterns enables smooth repositioning of the levitated particle throughout the interaction volume, allowing the voxel to trace arbitrary three-dimensional paths with high temporal coherence.

The levitated particle serves as the physical voxel. A lightweight bead typically 0.5–2 mm in diameter is suspended within the acoustic field and illuminated externally. Its motion, controlled by the phased array, forms the visible volumetric geometry. Because the voxel is a tangible object rather than an optical artifact, it provides high contrast, clear visibility, and stable behavior under a wide range of environmental conditions.

The illumination subsystem enhances voxel visibility and encodes geometric or physical attributes. A directed LED or laser diode illuminates the particle in synchronization with its position, ensuring crisp volumetric traces. Color, intensity, or temporal modulation can represent curvature, field strength, or SEFI layer identity, adding expressive depth to the rendered structures.

The geometric field engine based on the Single Entity Field Interpretation (SEFI) generates the voxel's trajectory. SEFI worldlines, curvature fields, and layered field constructs are converted into coordinate streams that define the particle's motion. These coordinates are compiled into phase patterns for the ultrasonic array, enabling the voxel to physically trace the structure of the underlying field model in real time.

This integrated architecture provides a direct physical embodiment of geometric computation. By combining established acoustic levitation techniques with a computational field model, the system renders dynamic three-dimensional structures in free space using a single controllable voxel. The following sections detail the acoustic levitation physics, dynamic voxel rendering model, SEFI integration, control mechanisms, and feasibility considerations.

3 Acoustic Levitation Physics

Acoustic levitation uses ultrasonic pressure fields to suspend small particles at stable positions within an open volume. When multiple transducers emit sound waves at the same frequency, their interference pattern forms regions of high and low acoustic pressure. A particle placed within this field experiences a net force that drives it toward pressure minima or maxima, depending on its size and acoustic contrast. By controlling the phase and amplitude of each transducer, these stable points known as acoustic nodes can be positioned precisely in three-dimensional space.

3.1 Standing Wave Formation

Ultrasonic transducers operating near 40 kHz generate sound waves with wavelengths on the order of 8–9 mm. When arranged in a phased array, the superposition of these waves creates standing wave patterns. The pressure distribution $p(\mathbf{r})$ at position $\mathbf{r}$ is determined by:

$$p(\mathbf{r}) = \sum_{i=1}^{N} A_i \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i), \tag{1}$$

where A_i is the amplitude of transducer i, $\mathbf{k}_i$ is its wave vector, and ϕ_i is its phase offset. Stable levitation points occur where the acoustic radiation force balances gravity and drag. These points can be shaped and repositioned by adjusting the phase offsets ϕ_i, allowing the levitated particle to be moved smoothly through the interaction volume.

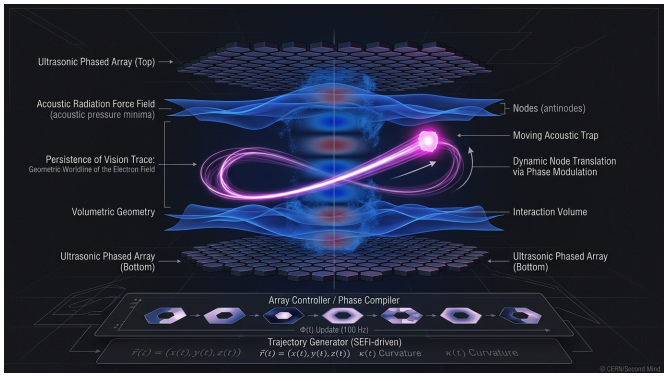


Figure 2: Schematic diagram of ultrasonic phased array pressure node translation driving a 3D parametric trajectory trace via persistence of vision.

3.2 Particle Requirements

The levitated voxel is a lightweight bead whose acoustic and optical properties support stable suspension and clear visibility. Suitable particles typically have:

- diameter: 0.5–2 mm,
- density: 20–200 kg/m^3,
- sufficient acoustic contrast to experience strong radiation forces,
- high optical reflectivity for crisp illumination.

Particles in this size range respond predictably to ultrasonic fields and remain stable under moderate environmental variation. Their small mass allows rapid repositioning, enabling dynamic trajectories suitable for volumetric rendering.

3.3 Dynamic Positioning

Dynamic voxel motion is achieved by continuously updating the phase pattern across the array. For a target coordinate $\mathbf{r}_t(t)$, the controller computes the required phase offsets:

$$\phi_i(t) = -\mathbf{k}_i \cdot \mathbf{r}_t(t). \tag{2}$$

This produces a moving pressure node that guides the particle along the desired trajectory. Smooth motion requires phase updates at 100–500 Hz, stable amplitude control, compensation for environmental drift, and avoidance of destructive interference patterns. When executed correctly, the particle can trace complex three-dimensional paths with high temporal coherence.

3.4 Stability Constraints

The stability of the levitated voxel depends on transducer alignment, phase accuracy, environmental noise, airflow, particle mass, and shape. Within typical operating conditions, the system maintains stable levitation across a volume of several cubic centimeters, sufficient for rendering worldlines, curvature fields, and layered SEFI constructs using dynamic voxel motion.

4 Dynamic Voxel Rendering

Dynamic voxel rendering is achieved by moving a single acoustically levitated particle along a controlled three-dimensional trajectory and illuminating it in synchronization with its position. When the particle moves rapidly enough, the human visual system integrates its motion into continuous volumetric curves and structures. This persistence-of-vision effect allows a single physical voxel to represent complex geometric forms without requiring multiple particles or a dense volumetric medium.

4.1 Single-Voxel Volumetric Tracing

The levitated particle acts as a point emitter or reflector whose position is updated continuously by the ultrasonic phased array. For a trajectory defined by:

$$\mathbf{r}(t) = (x(t), y(t), z(t)), \tag{3}$$

the voxel traces a visible curve in space when illuminated appropriately. The apparent geometry is determined by the spatial path of the particle, the speed of traversal, the illumination timing, and the viewer's persistence of vision. Smooth, high-frequency motion produces crisp volumetric traces, while slower motion emphasizes discrete point positions.



Figure 3: Close-up laboratory visualization of a single 0.5–2 mm illuminated particle tracing a continuous light trajectory under high-speed acoustic modulation.

4.2 Illumination Model

Illumination is synchronized with the particle's position to ensure that only the intended portions of the trajectory are visible. A directed LED or laser diode provides a narrow beam that highlights the voxel with minimal spillover. The illumination function $I(t)$ may encode geometric or physical attributes such as curvature, field strength, layer identity, entity membership, or temporal phase. For example, curvature magnitude $\kappa(t)$ can be mapped to brightness:

$$I(t) = f(\kappa(t)). \tag{4}$$

4.3 Temporal Multiplexing

Although the system uses a single physical voxel, it can represent multi-layer or multi-entity structures through temporal multiplexing. By cycling through multiple trajectories in rapid succession, the voxel renders layered SEFI constructs, multi-entity worldlines, stacked geometric fields, or composite volumetric shapes. Let $\mathbf{r}_i(t)$ denote the trajectory for layer or entity i; the system renders all layers by switching between trajectories at a frequency high enough to maintain perceptual continuity.

4.4 Volumetric Curve Quality

The clarity of rendered curves depends on voxel speed, illumination timing, acoustic stability, environmental conditions, and trajectory smoothness. Optimized trajectories minimize abrupt accelerations that could destabilize levitation. Illumination gating ensures that only valid positions contribute to the visible geometry, producing crisp, stable volumetric traces suitable for scientific visualization and geometric field exploration.

5 SEFI Geometry Integration

The volumetric rendering system derives its expressive capability from the integration of the Single Entity Field Interpretation (SEFI) with dynamic voxel motion. SEFI provides a geometric field model that defines worldlines, curvature distributions, and layered field constructs. These geometric outputs are translated directly into real-time trajectories for the levitated voxel, enabling the system to physically trace the structure of the underlying field model in free space.

5.1 Worldline Mapping

SEFI worldlines represent the geometric evolution of an entity's position or state within the field. Each worldline is defined as a parametric curve:

$$\mathbf{r}(t) = (x(t), y(t), z(t)), \tag{5}$$

which the voxel traces through dynamic acoustic positioning. Because the voxel is a physical particle, the rendered worldline becomes a tangible, observable curve suspended in space.

5.2 Curvature Visualization

Curvature of a worldline is given by:

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}. \tag{6}$$

Curvature can be encoded visually through illumination intensity, color modulation, trajectory speed, or temporal gating, allowing the viewer to perceive geometric field behavior directly through the voxel's motion and appearance.

5.3 Layered Field Constructs

SEFI defines multiple geometric layers that describe different aspects of field behavior. These layers can be rendered using temporal multiplexing, where the voxel cycles through multiple trajectories at high frequency. Layer identity can be encoded through color, brightness, traversal order, or modulation frequency, enabling visualization of SEFI's multi-layer field constructs.

5.4 Entity and Interaction Representation

Multiple SEFI entities can be represented by cycling through their worldlines or by rendering interaction paths that describe relationships between entities. Interaction geometry such as convergence, divergence, or coupling can be visualized through shared trajectory segments, synchronized illumination patterns, alternating voxel motion, or curvature-based emphasis.

5.5 Observer and Operator Geometry

SEFI includes geometric constructs that describe the observer's frame or operator perspective. These can be rendered as reference curves or volumetric scaffolds that contextualize entity motion. The voxel can trace coordinate axes, reference frames, operator paths, or geometric anchors.

6 Control Architecture

The control architecture coordinates the geometric field engine, ultrasonic phased array, and illumination subsystem to produce stable, expressive volumetric rendering. It is designed as a modular pipeline in which each stage transforms geometric information into physical behavior.

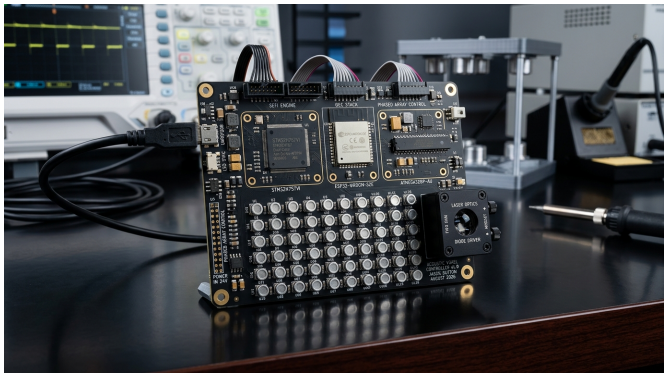


Figure 4: Custom embedded transducer driver board and laser diode optics assembly configured for high-frequency phase compilation and feedback control.

6.1 SEFI Engine to Trajectory Generator

The SEFI engine computes worldlines, curvature fields, and layered field constructs, producing $\mathbf{r}(t) = (x(t), y(t), z(t))$. The trajectory generator smooths discontinuities, enforces acceleration limits, resamples coordinates at a fixed control frequency, and prepares multiplexed trajectories.

6.2 Trajectory to Phase Pattern Compiler

The phase pattern compiler converts each target coordinate into phase offsets:

$$\phi_i(t) = -\mathbf{k}_i \cdot \mathbf{r}(t). \tag{7}$$

This is performed at high frequency to maintain smooth voxel motion, with amplitude normalization and stability constraints.

6.3 Ultrasonic Array Controller

The array controller drives each transducer with the specified phase and amplitude, forming the pressure node that levitates the particle. It provides synchronized timing, drift compensation, and safety monitoring.

6.4 Illumination Synchronization

The illumination subsystem receives the same coordinate stream and activates the light source only when the voxel occupies the correct spatial position. Illumination parameters encode SEFI attributes such as curvature magnitude, layer identity, interaction strength, or temporal phase.

6.5 Feedback and Stability Monitoring

Optional feedback mechanisms include optical tracking of voxel position, acoustic field monitoring, environmental sensing, and adaptive phase correction.

6.6 Integrated Control Loop

1. SEFI engine generates geometric coordinates.
2. Trajectory generator refines and multiplexes paths.
3. Phase compiler converts coordinates into acoustic phase patterns.
4. Array controller forms and moves the levitation node.
5. Illumination controller lights the voxel in sync with its motion.
6. Feedback systems maintain stability and compensate for drift.

7 Safety Envelope

The volumetric rendering system operates within a well-defined safety envelope that ensures reliable performance and safe interaction. Acoustic levitation at ultrasonic frequencies is non-ionizing and non-thermal, and illumination remains within eye-safe limits.

7.1 Acoustic Safety

The array operates near 40 kHz, above human auditory perception. Acoustic output is limited to safe pressure levels, with monitoring of transducer load and thermal behavior.

7.2 Optical Safety

Directed LEDs or low-power laser diodes are used with controlled intensity, beam divergence, and gating, keeping illumination within safe exposure thresholds.

7.3 Particle Safety and Containment

The levitated voxel is a lightweight bead with low mass. Particle size and material are chosen to avoid inhalation risk and toxicity, with optional containment structures for laboratory environments.

7.4 Environmental Stability

Environmental factors such as airflow and temperature gradients are mitigated through sensing, shielding, and stable mounting.

7.5 Operational Safeguards

Safeguards include thermal monitoring, automatic shutdown on unsafe acoustic output, illumination gating, trajectory validation, and controller watchdogs.

8 Feasibility Analysis

The system is low-cost, technically accessible, and based on established techniques.

8.1 Hardware Cost

Key components:

- Ultrasonic phased array: \$80–\$300
- Microcontroller: \$10–\$40
- Illumination subsystem: \$5–\$20
- Levitated particles: negligible cost
- Mechanical frame: \$10–\$30

Total system cost: approximately \$200–\$400.

8.2 Engineering Feasibility

All core technologies—acoustic levitation, dynamic positioning, persistence-of-vision rendering, synchronized illumination—are established and documented.

8.3 Software and Control Complexity

The control architecture is modular and implementable using standard embedded systems and computational geometry techniques.

8.4 Technical Risks and Mitigations

Risks such as levitation stability, trajectory smoothness, environmental sensitivity, and illumination precision are manageable through engineering controls.

8.5 Development Path

1. Static levitation
2. Slow dynamic positioning
3. Full trajectory tracing
4. Illumination synchronization
5. SEFI-driven rendering
6. Layered field visualization

9 Applications for Microsoft AI Companions & Spatial Systems

The integration of SEFI-driven trajectory engines with acoustic levitation unlocks direct applications for interactive physical AI platforms, spatial audio research, and physical Copilot interfaces.



Figure 5: Interactive spatial application showing mid-air tactile particle manipulation and free-space visual anchor integration in an experimental laboratory environment.

9.1 Acoustic Field Visualization & Physical AI Embodiment

Serves as a physical embodiment of SEFI, rendering acoustic energy distributions, wave interference patterns, curvature, and multi-layered field structures in mid-air. It provides an experimental benchmark for testing physical AI indicators, standing wave node stability, transducer phase control, and dynamic spatial responses.

9.2 Mixed Reality and Spatial Interaction

Acts as a physical free-space cursor or spatial sound anchor, combining physical visual cues with directional audio transducers for physical Copilot and mixed-reality UX paradigms.

10 Conclusion

This paper presents a volumetric rendering system that uses acoustic levitation to position and manipulate a single illuminated particle as a dynamic voxel. By integrating established ultrasonic control techniques with a geometric field model, the system produces expressive three-dimensional structures through persistence of vision. The Single Entity Field Interpretation (SEFI) provides the mathematical foundation for generating worldlines, curvature fields, and layered constructs, which are translated directly into real-time voxel trajectories.

The system's architecture is compact, low-cost, and grounded in well-validated physical principles. Its modular control pipeline enables stable and precise voxel motion, and the resulting volumetric traces offer a clear method for visualizing geometric, acoustic, and physical AI interaction phenomena. By demonstrating that complex geometric structures can be rendered using a single levitated voxel, the system establishes a foundation for future physical AI companion hardware and spatial display technologies.